

# Allometric calibration bias in tropical forest carbon

Architecture-informed measurement, LiDAR dependence and annual greenhouse-gas attribution

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## Abstract

Forest-carbon estimation links a biological reference, a spatial prediction and an annual carbon account. We assemble these operations into a framework that separates accuracy of the reference from precision of its spatial reproduction. Conditional-expectation identities show that an ideal LiDAR predictor retains the conditional bias of its training labels; a population-transfer identity isolates the effect of changing architectural composition. Stock-weighted covariance determines when a carbon-stock correction transfers unchanged into annual release. The paired-account and finite-increment results distinguish common-scale cancellation from differential amplification of sources and removals. A published-digit analysis resolves two class assignments and characterizes a shared normalization in the initial 723.97 PgC arXiv application. A stem–branch volume decomposition turns its high-density class into an explicit test of joint basal area, height, material density and organ allocation. Illustrative low-cylinder scenarios fall below that target, whereas the cited mean crown allocation does not establish a class-wide physical ceiling. Three conditional accounting outcomes connect a supported forest-source correction to a larger total, increased forest attribution within a retained total, or joint source–removal revision. A matched historical scenario demonstrates a largest-land-group ranking with other absolute contributions retained; independent constraints govern empirical application. These results direct calibration towards independently measured organ volumes and tissue carbon, representative spatial weights and disturbance-specific observations. The framework identifies what must change in an estimator and what must change in an annual account, while keeping empirical support for the allometric-bias mechanism distinct from validation of the initial map’s magnitude.

**Keywords:** allometric calibration bias; tropical forest carbon; tree architecture; lidar; reference-label bias; biological volume; tissue carbon; deforestation and degradation; stock-to-flux transfer.

## 1 Introduction

The reliability of forest-carbon accounting begins with the biological quantity assigned to trees and stands, not with the spatial precision of the map used to distribute it. Allometry-based carbon labels may omit or misrepresent architecture and organ-level composition even when tree position, diameter or canopy height is accurately measured. When those labels calibrate a remote-sensing product, increased coverage can reproduce the same carbon-scale error over a larger area. A sensor measures properties of the forest; the conversion of those properties into biomass and carbon remains a separate inference.

This article examines the propagation of structural and compositional bias in allometry-based forest-carbon estimation into spatial products and annual greenhouse-gas accounts. Architecture-informed measurement and the initial pantropical reassessment provide the methodological basis and application [1, 2]. The later NEBIOT essay discusses their interpretive implications [3]. Their central mechanism is that the actual material of stems, branches and other organs can differ from the material represented by a

generalized prediction. Measuring that architecture is consequently a way to diagnose allometric bias, rather than an unrelated explanation for the resulting discrepancy.

The contribution is the assembly of the measurement, calibration and accounting operators. The same weighting principle exposes reference-label inheritance, stock-to-flux covariance and different source, removal, gain and loss corrections; the relevant weights and biological quantities change at each stage. The classical identities are used to specify observable mechanisms and discriminating tests. Source-table diagnostics and detailed material scenarios support that argument in the technical supplement.

**Status of the initial estimate.** The 723.97 PgC value is the initial architecture-informed estimate in arXiv:1508.03667, first posted on 14 August 2015 and revised on 17 June 2016; it is a preprint result, not a peer-reviewed inventory determination. It covers the mapped pantropical aboveground-vegetation domain excluding the Australian tropics, with nominal map epoch 2007–2008 [1, pp. 23–24]. The author is developing successor calibrations; this article supplies neither a replacement map nor a newly validated total. Its analytical results are independent of that initial numerical value. Independent empirical verification of the class-5 mean remains outstanding.

The evidence synthesis is purposive rather than a meta-analysis. The author developed the architecture-informed methodology underlying the cited preprint. Original measurements, source-reported method comparisons and analytical identities are identified separately; no synthetic stand is represented as a new field observation. Figures 1–3 connect the reported inventory, the forest-carbon interpretation and the two calibration pathways. Figure 2 presents the numerical second interpretation with a questioned total, a double-headed arrow and a conditional largest-sector comparison, while keeping the standing-stock record separate from annual flows. The sector–activity–gas layouts of the first two figures adapt the WRI and Ecofys visual convention [4, 5]; intermediate activity nodes are conceptual pathways, not an alternative official sector classification.

## 2 Allometric calibration bias as a measurement problem

### 2.1 Architecture as a mechanism of allometric calibration bias

Aboveground biomass allometry predicts dry mass from a selected set of tree attributes. A representative formulation is  $\hat{B}_A = g_\theta(D, H, \rho)$ , where diameter, height and density provide informative but incomplete descriptions of a tree. An architecture-informed estimate instead resolves the material in its organs. Two individuals sharing the predictors of a reduced-input relationship can differ in crown extent, branch hierarchy, buttress form, damage and allocation among organs. A population-average prediction need not recover either individual’s mass exactly. Systematic differences between the calibration population and the forest being assessed can then turn unresolved individual variation into aggregate bias.

Underrepresented branch or stem material is a physical mechanism of allometric underestimation. A systematic discrepancy can therefore arise within the woody material already assigned to an aboveground-biomass target. The relevant comparison is between predicted and actual material in the same woody and nonwoody compartments. This distinction also separates target completeness from estimator accuracy: an equation may be intended to predict total aboveground biomass and still represent the distribution of that biomass imperfectly.

Goodman et al.’s 51-tree sample provides direct evidence of the architectural mechanism: mean crown allocation was  $44\% \pm 2\%$  of aboveground biomass, with individual fractions of 14–71%, and the tested height-inclusive pantropical equations underestimated the sample total by 11–14% [6]. Their crown component includes branches and attached foliage/reproductive material; ten stems used sectional volume–density estimation rather than complete weighing. Crown measurements improved prediction. A broader, partly overlapping synthesis of 673 harvested trees found crown fractions of approximately 3–88%

and a size-dependent allocation pattern [7]. These are empirical distributions and estimator comparisons, not universal limits on crown volume or class-average carbon.

## 2.2 Conditional representation and transfer between populations

Let  $X$  contain the recorded predictors and let  $Z$  contain additional architecture and composition. Write true biomass as a square-integrable function  $B = \phi(X, Z)$  on the defined pool. For an ideal squared-error predictor calibrated in population  $P$ ,  $g_P(x) = \mathbb{E}_P[B \mid X = x]$ . Even this ideal predictor has residual variation

$$\mathbb{E}_P[(B - g_P(X))^2] = \mathbb{E}_P[\text{Var}_P(B \mid X)]. \quad (1)$$

The conditional-mean property gives zero mean prediction error in  $P$ ; unresolved variation alone is therefore not a proof of population bias. Its role changes when the conditional distribution of architecture or composition changes. For a target population  $Q$  on overlapping predictor support,

$$g_P(x) - \mathbb{E}_Q[B \mid X = x] = \int \phi(x, z) \{dP(z \mid x) - dQ(z \mid x)\}. \quad (2)$$

The expression is exact when the same biological response function is used and the expectations exist. Finite-sample fitting, predictor measurement errors and a misspecified functional form add further errors. The formula identifies a mechanism: using the same diameter and height range does not establish that the unrecorded architecture is distributed in the same way. A downward error requires the conditional biomass in the target population to exceed the calibration expectation, rather than being assumed from heterogeneity alone.

A synthetic example illustrates the distinction. Suppose trees share the recorded predictors but have two architectural states with biomass 100 and 200 arbitrary units. Equal calibration frequencies give a prediction of 150. If the target frequencies are 0.2 and 0.8, the target mean is 180 and the transferred prediction is 30 units lower, a 16.7% shortfall. Increasing the number of trees assessed with the unchanged rule does not alter that expected discrepancy. These are analytical values, not measured tropical forest corrections; the executable example is provided in the numerical companion.

Homogeneous species, ages and management can narrow the conditional variation that a fitted model must represent, but homogeneity is not the criterion defining the validity of allometry. Chave et al. evaluated equations against 4,004 harvested trees from 58 tropical sites and reported broad predictive performance [8]. Applicability is therefore established by validated transfer across architecture and composition, rather than by a crop-versus-natural-forest classification. Crop and plantation relationships require calibration checks, while natural-forest relationships can be useful within independently supported domains. Treating either as an independently measured carbon scale without testing that transfer is the methodological error addressed here. What heterogeneous tropical forests particularly demand is explicit testing of architecture, composition and disturbance states that reduced predictors may fail to distinguish.

## 2.3 The working hypothesis and its observable consequences

The central hypothesis is that structural and compositional underrepresentation in the allometric calibration of some tropical forest populations produces consequential errors in carbon stocks and the fluxes derived from them. The architecture-informed reassessment supplies the methodological basis and application [1, 2]. Its reported discrepancies motivate measurement at the level where the error is proposed to arise: organs, trees and vegetation populations, followed by the spatial and temporal operations used in accounting.

The hypothesis has observable consequences. At matched tree and plot supports, independent reconstructed material and tissue carbon should explain signed residuals of the allometric estimator, with residual structure related to architectural or compositional attributes omitted from that estimator. Replacing

biased reference labels should alter remote-sensing carbon predictions even when the sensor observations are unchanged. Disturbance-weighted residuals should then predict the correction to the source population more directly than the unweighted mean correction across intact forest. These predictions admit localized support, mixed signs or failure; their measurement does not require treating conventional inventory agreement as the definition of correctness.

### 3 Why LiDAR coverage does not by itself resolve carbon-scale bias

#### 3.1 Sensor observations, calibration labels and independent reference

LiDAR supplies geometric information through laser ranging, return waveforms or point clouds. Biomass and carbon require additional material or empirical relationships. The crucial distinction is therefore the *inference pathway*, not the sensor name. A height-based biomass regression calibrated on allometric plot estimates remains dependent on those reference estimates. Detailed laser reconstruction of material volumes combined with independently measured density and carbon fractions can instead reduce that dependence. Both pathways have uncertainties, but they are not the same measurement procedure.

The documented GEDI Level-4A Version-2 pathway relates waveform relative-height metrics to field-plot estimates of aboveground biomass density, using regional and plant-functional-type models. Its guide describes geographic cross-validation and flags observations outside predictor or response support [9]. These controls address transfer and prediction errors; validation against reference labels alone cannot expose a systematic error common to the labels. This observation is not an estimate of the bias of a particular GEDI release. It identifies a component of reference uncertainty that must be established from the underlying field methods.

Conversely, Calders et al. reconstructed woody volume from terrestrial laser scanning and combined it with basic density in an Australian eucalypt forest. Against 65 harvested reference trees, the tested allometric totals were 29.85–36.57% low, whereas the TLS reconstruction was 9.68% high [10]. This demonstrates a route through direct geometric reconstruction rather than the same biomass allometry; it also demonstrates that laser-derived reconstruction is not error-free. Registration, occlusion, segmentation, branch geometry, leaf–wood separation and material-property sampling require their own checks. Thus increased sensing coverage cannot cure a biased calibration merely by being extensive, while appropriately used LiDAR can be part of the architectural solution.

**Table 1:** Three distinct measurement pathways. The same sensor can participate in different calibration chains; independence is an attribute of the chain, not of spatial resolution alone.

| Pathway                                           | Carbon information                                              | Principal calibration question                                                      |
|---------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Field predictors plus biomass allometry           | Empirical mass prediction from recorded attributes              | Does calibration represent the target architecture, composition and disturbance?    |
| Remote structure plus allometric reference labels | Spatial prediction of the reference biomass/carbon scale        | Does validation test true material, or repeat the label-generating estimator?       |
| Organ reconstruction plus material analysis       | Volume, basic density and tissue carbon, then spatial extension | Are the volumes and properties independently measured and spatially representative? |

#### 3.2 Segmented terrestrial reconstruction and plot-scale processing

Disney et al. describe how terrestrial laser scanning (TLS) resolves tree form and woody material independently of a conventional biomass scaling equation [11]. The relevant sequence is registration and leaf–wood classification, individual-tree segmentation, quantitative structure modelling, and conversion

of reconstructed volume using material properties. Segmentation alone supplies neither biomass nor elemental carbon. This pathway can supply new reference measurements, rather than merely predict the allometric labels used by an existing map; geometry, density and chemistry nevertheless retain separate uncertainties.

A direct regional example is Calders et al.’s Wytham Woods study, coauthored by Disney [12]. Across 1.4 ha and 815 live trees, TLS-based aboveground biomass was  $409.9 \text{ Mg ha}^{-1}$ , 1.77 times the estimate from the comparison allometries. This was a temperate stand reconstruction using quantitative structure models and assigned species-density information, not complete independent mass weighing. It supports the possibility of substantial reference-calibration bias and a route to reducing it; it does not establish a pantropical multiplier or validate  $723.97 \text{ PgC}$ .

Wilkes, Disney and colleagues’ TLS2trees pipeline addresses automated, plot-scale segmentation [13]. Its 7.5-ha, ten-plot demonstration covered seven forest types and segmented 10,557 trees; 1,053 were paired with manually segmented reference trees. Their Table 2 reports instance-segmentation wall times of 16 and 18 h for the one-hectare NOU and MLA datasets, with maximum RAM per tile of 51.6 and 129.8 GB, respectively. These are implementation-specific benchmarks, not universal per-hectare costs. They substantiate the computational bottleneck while showing why tiled, parallel processing is useful. Errors in tree separation and small-tree recovery remain relevant to complete plot biomass. Thus scalable sensing, scalable segmentation and independently accurate material calibration are three distinct requirements.

### 3.3 Calibration-label inheritance

Let  $C$  denote true carbon on a fixed pool and support,  $Y = C + b + \varepsilon$  the calibration label, and  $L$  the remote-sensing predictors. Here  $b$  is the label’s signed systematic discrepancy and  $\mathbb{E}_P[\varepsilon | L] = 0$ . In the population limit of squared-error prediction, define  $m_Y(l) = \mathbb{E}_P[Y | L = l]$  and  $m_C(l) = \mathbb{E}_P[C | L = l]$ . Linearity of conditional expectation gives

$$m_Y(l) - m_C(l) = \mathbb{E}_P[b | L = l]. \quad (3)$$

This is an exact conditional-mean identity, not a claim of a new statistical theorem or of universal consistency for a particular learning algorithm. If a predictor accurately reproduces  $Y$ , its discrepancy from  $C$  can still equal the conditional label bias. An average-zero label error independent of  $L$  behaves differently: it contributes noise but no conditional-mean displacement in this ideal case. A target-population shift adds a second component,

$$m_Y(l) - \mathbb{E}_Q[C | L = l] = \mathbb{E}_P[b | L = l] + \{\mathbb{E}_P[C | L = l] - \mathbb{E}_Q[C | L = l]\}. \quad (4)$$

Separate reference-mass validation and transfer diagnostics are consequently required. Where predictors cannot distinguish structurally different trees or plots, accurate ranging cannot supply the omitted material information through a response label that already misrepresents it.

A complementary shared-error example makes the coverage distinction explicit. If  $Y_i = C_i + b_0 + \varepsilon_i$ , with a fixed shared offset  $b_0$  and independent zero-mean errors of variance  $\sigma^2$ , then the mean error has mean-squared value

$$\mathbb{E}[(\bar{Y}_n - \bar{C}_n)^2] = b_0^2 + \sigma^2/n. \quad (5)$$

More samples reduce the independent component, not the squared shared offset. Correlated errors require their full covariance instead of  $\sigma^2/n$ . Actual wall-to-wall pixels are not independent reference measurements, and a denser point cloud should not be equated with  $n$  independent carbon calibrations. The result separates spatial detail, statistical precision and absolute carbon accuracy without denying the value of better sensing.

### 3.4 Replacing the reference rather than merely refining the map

The constructive response is to calibrate the carbon scale from independently checked architecture and tissue properties, while using remote sensing to extend those labels through a documented support domain. Randomly splitting outputs from one label-generating model only tests prediction of that model. Complete site and vegetation holdouts, independent material references, and signed residuals against both allometric and architecture-based estimates provide stronger evidence. The same safeguard applies when an architectural model supplies training labels: geometry must be validated before label replication is interpreted as physical accuracy.

Calibration improvements may change the regional or pantropical point estimate without invalidating the measurement argument. The 723.97 PgC estimate reported on arXiv remains the reported application; it is neither recalculated from a LiDAR product here nor made the value that future calibration must reproduce. The methodological objective is the fidelity of the material-to-carbon chain and the consequences of replacing biased reference quantities. Figure 3 summarizes the two routes from field information to spatial products and annual accounts.

## 4 Architecture-informed measurement and the pantropical application

### 4.1 The architectural measurement chain and its physical tests

A dimensionally explicit tree-carbon representation is

$$C_j = \sum_o V_{jo} \rho_{jo}^b f_{jo}^C, \quad (6)$$

where non-overlapping compartments  $o$  have green material volume  $V$ , basic density  $\rho^b$  defined as oven-dry mass per green volume, and dry-mass carbon fraction  $f^C$ . With consistent units, the product is carbon mass. Hollow space, bark, leaves and branches must not be double counted; roots do not belong in an aboveground-only target. Moisture conventions must be matched rather than multiplying hydrated mass by a dry-mass carbon fraction.

Pantropical allometries can incorporate diameter, height and wood density [8], yet trees sharing those predictors can differ in crown architecture, branch allocation, hollow volume and tissue composition. A relationship based on the reduced predictors can describe their calibration-population average without recovering every individual's material distribution. Architecture-informed reconstruction resolves additional quantities, so its validity is not conditional on reproducing an allometric prediction. Tissue chemistry provides a separate measurement axis: a biomass correction and a carbon-fraction correction may have different signs [14]. The IPCC tropical aboveground default of 0.47 is an accounting parameter for specified applications, not a substitute for measured organ fractions [15].

NEB002 combines measured architecture, numerical integration or voxelization, material density and elemental carbon measurements; NEB001 links vegetation identity and environmental structure to spatial mapping [2]. Neural networks extend the resulting carbon estimates. The physical inputs therefore precede the statistical extension. Validation is assessed at each step: representation of measured material, assignment of density and chemistry, prediction beyond the reconstructed individuals, and integration across vegetation types. A model's ability to reproduce its training targets is relevant to the prediction step but does not summarize all the evidence supporting those targets.

The preprint reports 221 reconstructed individuals from 127 species (p. 6), and 2,232 tissue samples from 186 individuals processed with a LECO CHN-600 analyzer (pp. 16–17). Chemical quality control used reference compounds at ten-sample intervals. Its physical volume comparison used 10 individuals of *Aphelandra pulcherrima*, 1.54–2.9 m tall, with cutting, weighing and water displacement (p. 25 and Fig. 11). The comparison is summarized in Table 2; the measured material and the geometrical predictions

are distinct observations [1].

**Table 2:** Reported physical-volume validation in the original preprint [1, p. 25]. The same 10 *Aphelandra pulcherrima* individuals provide the reference and reconstruction comparison. Mean volumes are source-reported; the final column gives the non-negative shortfall from the water-displacement reference, calculated from the displayed means and rounded to one decimal place. All three reconstructions fall below the reference, with the 1-mm voxel grid showing the largest shortfall. It is a difference of sample means, not an individual-error bound or a pantropical confidence interval.

| Method                         | Mean volume (cm <sup>3</sup> ) | Shortfall from reference (%) |
|--------------------------------|--------------------------------|------------------------------|
| Water displacement (reference) | 1,240.00                       | 0.0                          |
| Numerical integration          | 1,217.76                       | 1.8                          |
| Voxelization, 1-cm resolution  | 1,211.00                       | 2.3                          |
| Voxelization, 1-mm resolution  | 1,189.92                       | 4.0                          |

This is physical validation of volume at the tested scale, alongside laboratory measurements of tissue carbon; it cannot be reduced to neural-network self-consistency. Its extension to different sizes and architectures is a distinct validation question. Numerical accuracy in integrating a specified mesh and accuracy of the mesh as a representation of a tree also remain separate quantities. Independent geometry, density and chemistry checks address these steps without requiring agreement with a conventional biomass equation.

The regional chain likewise precedes the pantropical total. The preprint reports network-based carbon estimates for more than 6,000 individuals in 27 vegetation types and a wider regional comparison across 54 vegetation types. Their reported average carbon densities were 177.03 MgC ha<sup>-1</sup> for the architecture-informed method and 61.41 for the allometric comparison. The pantropical calibration used regional carbon targets with the carbon map published by Baccini et al. in 2012 and the height map published by Simard et al. in 2011 [1, pp. 22–24]. Predictor and calibration dates are distinguished in Table 5. A conventional map used as a predictor is not thereby accepted as the true carbon target; shared predictors can nevertheless induce dependence between products. The complete chain is measured organs → reconstructed trees → vegetation-specific carbon estimates → spatial calibration, rather than a post hoc multiplication of an accepted global inventory.

## 4.2 Calibration support and material requirements

The reported regional mean of 177.03 MgC ha<sup>-1</sup> is not the largest calibration response. Page 23 of the arXiv source states predictor ranges of 0–190 MgC ha<sup>-1</sup> for the Baccini map and 0–38 m for Simard height, with regional carbon responses spanning 0–523 MgC ha<sup>-1</sup> [1]. The class-5 lower edge, 478, is within that marginal response range; its mean and maximum exceed the endpoint. Thus some predictions extend beyond the recorded response range, but the mean’s exceedance does not place every class-5 cell outside it or establish extrapolation in the multivariate predictor domain. Joint predictor support and signed error require record-level labels, model inputs and independently held-out observations.

If the 478–605 MgC ha<sup>-1</sup> interval is the implemented raster definition and the ratio 536.33 is its area-weighted mean on the stated mask, at least 16.25% of class-5 area (86.4 million hectares) exceeds the page-23 response endpoint of 523. Capping only that exceedance at 523 removes between 7.08 and 20.01 PgC, or less than 2.77% of the reported pantropical total. The printed-mean convention gives a minimum bound of approximately 9.82% (52.2 million hectares) and a 4.28–18.20 PgC clipping change on the same illustrative area. Supplementary Note S1 derives the sharp moment bounds and shows that the affected area can nevertheless be as large as the whole class. These bounds concern a specified numerical cap, not the magnitude of calibration error: biased in-range labels, joint predictor extrapolation and errors in other classes can persist after capping. Page 22 separately reports a regional value of 523.4 MgC ha<sup>-1</sup>;

the archive supplies this nearby-endpoint sensitivity rather than treating the rounded page-23 range as a recovered exact label maximum [1].

If every mapped compartment's carbon fraction lies in the preprint's measured tissue range 0.4111–0.4999, the class edge implies at least about 956 Mg dry biomass ha<sup>-1</sup>, and the ratio-based class mean implies about 1,073–1,305 Mg ha<sup>-1</sup>. These are conditional bounds from the reported tissue range, without choosing 0.47 as a uniform fraction; they are not a reconstructed biomass map or a demonstrated global tissue-fraction bound.

The initial map's standing and successor-calibration programme are specified in the [status box](#). Alternative totals also depend on their domains: Feldpausch et al.'s 285.1 PgC exceeds Baccini's 228.7 on a different mask, while representing a downward height correction from 320.5 on Feldpausch's own unchanged mask; Avitabile et al.'s fusion also lowers its input estimates [16–18].

**Interpreting the high-density requirement.** At the illustrative carbon fraction 0.47 and basic density 0.60 Mg m<sup>-3</sup>, the ratio-based class mean requires approximately 1,900 m<sup>3</sup> ha<sup>-1</sup> of wood when the separate remainder is zero. A stem–branch decomposition in Supplementary Note S1 shows that a scenario with stem form ratio 0.4–0.6, branch fraction 0.44 and equal compartment densities gives whole-wood ratios of 0.71–1.07. Combined with basal area 30–40 m<sup>2</sup> ha<sup>-1</sup> and weighted height 30 m, it yields approximately 181–363 MgC ha<sup>-1</sup>, below the class target. At basal area 50 and weighted height 40, however, the same parameter range gives approximately 403–604 MgC ha<sup>-1</sup>, spanning the target. These are sensitivity configurations, not observed class means or established plot maxima. The published crown fractions are sample distributions; their mean cannot serve as an upper bound on the required cylinder-weighted volume ratio. The comparison identifies the joint structural conditions needed to attain the mapped density, whose class-average empirical support remains unverified. Independent joint upper bounds could reject the class mean even without the original raster; the cited studies do not establish such a bound for its population. Representative geometry, material and sampling records remain necessary to validate that mean.

**Acquisition and discretization provenance.** The preprint describes field dimensions, photographs, scientific drawings and procedural reconstruction with SpeedTree and Xfrog [1, pp. 6–10]. The accessible record does not assign each of the 221 trees to a mutually exclusive capture route or identify an independent geometric check for every tree. Modern acquisition options cannot be assigned retrospectively to those records. The companion capture-route schema separates observed inputs, procedurally generated organs, volume algorithm, mesh identifiers and independent checks; its row set is deliberately empty until primary records are supplied.

The 1-mm voxel mean falls farther from both numerical integration and displacement than the 1-cm mean. A non-watertight mesh combined with an inside/outside rule is one testable cause; changed meshes, scale or origin, different occupied-cell rules and averaging also require checking. A consistent sequence on one closed mesh should converge as the grid is refined, although successive errors need not decrease monotonically. The aggregate means do not identify the cause; the original mesh and paired inclusion settings distinguish these mechanisms.

### 4.3 Biological material is not merchantable timber

The physical target is the material in the defined biological compartments, irrespective of saleability. For disjoint woody components on the same live aboveground boundary,

$$V_W = V_{\text{merchantable}} + V_{\text{nonmerchantable}}. \quad (7)$$

Neither a minimum commercial top diameter nor timber quality removes existing carbon. Stem tops, branches, twigs, buttresses and noncommercial individuals must enter the target where present, with

foliage or other nonwoody organs treated separately. Cavities contain no woody material; carbon remaining in degraded tissue must be represented with appropriate material properties and live/dead assignment rather than being excluded for lack of timber value. An outer crown envelope contains air and is not a material volume.

IPCC Box 4.2 explicitly distinguishes merchantable volume from total biomass and describes expansion of timber-volume records to additional components [15]. This is a definition of the measurement boundary, not validation of any default expansion factor. Direct biomass allometries may also target whole-tree aboveground dry mass, so it would be incorrect to equate every allometric output with saleable timber. The structural issue is whether the selected predictor or expansion actually represents the nonmerchantable material of the target vegetation.

For measured volume  $V$  and basic density  $\rho^b$  defined as oven-dry mass per corresponding green volume,  $V\rho^b$  estimates dry mass; multiplying by the matched dry-mass carbon fraction gives carbon. Hydrated mass, green volume, dry volume and elemental-carbon fraction cannot be interchanged. Density and carbon fractions vary by organ and sample, and their products must be aggregated with the relevant material weights. A single assumed fraction is useful for an explicitly marked sensitivity calculation, not as a retroactive description of a method that measured tissue carbon directly.

Consequently, a hypothetical large unmeasured foliage remainder is not required to explain underestimated woody biomass. The disagreement can concern the woody material itself. Component completeness, prediction error within already included components, and spatial selection are distinct diagnostic questions. None should be used to replace the allometric-bias mechanism with an unsupported assumption that the conventional woody subtotal is exact.

#### 4.4 Intermediate stems, abundance and forest degradation

Forest-carbon estimation and release accounting concern the full sampled size distribution. The architectural preprint compares organ biomass across diameters of 1–20 cm in Figure 6, extends its comparison to larger diameters in Figure 7, and presents carbon estimates for more than 6,000 individuals in Figure 8 [1, pp. 20–22, Figs. 6–8]. Its Table 1 contains, for example, aboveground-biomass estimates of 104.82 kg for *Inga* sp. 10 (15.24 cm DBH), 109.84 kg for *Pouteria* sp. 4 (11.58 cm DBH), and 115.74 kg for *Virola elongata* (14.63 cm DBH). These are biomass outputs, whereas Figure 8 uses kilograms of carbon. The role of intermediate stems is therefore part of the original measurement domain, rather than an extrapolation from a large-tree validation sample.

A complementary stand-structure result appears in Vásquez and Arellano [19, Table 170, p. 942]. In the *Acalypha* sp.–*Guazuma ulmifolia* community, the 10–30 cm class contributes a reported 38.97% of aboveground biomass and 38.79% of carbon. Dividing its 2.88 tonnes of biomass by 30 stems per 0.05 ha gives 96 kg biomass per stem (Table 3). This ratio of displayed class aggregates illustrates an approximately 100 kg mean, not an assertion that every stem has that mass. Classes II and III together account for 62.71% of that community’s biomass. “Intermediate” here follows the local diameter classes; it is not a universal mass threshold.

The 2012 chapter estimated biomass using selected Chave et al. relationships and measured tissue-carbon fractions; the later architectural reconstruction is a distinct method. Its *Acalypha*–*Guazuma* community total is 7.39 Mg per 0.05 ha, or 147.8 Mg ha<sup>-1</sup>, placing the abundance example in a comparatively low-biomass stand within that study [19, Table 170, p. 942]. It illustrates a size-class distribution, rather than supplying independent support for the class-5 density. The near-100 kg architectural records instead establish the size coverage of that method. Together with stands where larger classes dominate, the examples motivate vegetation-specific population weighting.

**Table 3:** A published intermediate-stem example: the *Acalypha* sp.–*Guazuma ulmifolia* community in Córdoba [19, Table 170, p. 942]. Class II is 10–30 cm DBH in the table. Inputs and percentage shares are source-reported; means per stem are derived from the displayed aggregates. The chapter’s abstract gives a different lower-class interval, so the table’s explicit bin definition is used here.

| Quantity                           | Value | Basis    |
|------------------------------------|-------|----------|
| Stems per 0.05 ha                  | 30    | Reported |
| Biomass (tonnes per 0.05 ha)       | 2.88  | Reported |
| Carbon (tonnes C per 0.05 ha)      | 1.30  | Reported |
| Share of community biomass (%)     | 38.97 | Reported |
| Share of community carbon (%)      | 38.79 | Reported |
| Mean biomass per stem (kg)         | 96.00 | Derived  |
| Mean carbon per stem (kgC)         | 43.33 | Derived  |
| Classes II + III biomass share (%) | 62.71 | Reported |

For disjoint classes  $h$ , let  $n_h$  be stem abundance and  $C_j$  the carbon of an individual. Then

$$C_h = \sum_{j \in h} C_j = n_h \bar{C}_h, \quad F_h(t) = \frac{44}{12} \sum_{\tau \leq t} \sum_{j \in \mathcal{D}_{h,\tau}} \sum_o C_{jo,\tau} d_{jo,\tau} k_{jo,\tau}(t - \tau).$$

Here  $\mathcal{D}_{h,\tau}$  indexes affected stems,  $o$  indexes disjoint organs,  $0 \leq d_{jo,\tau} \leq 1$  is the affected fraction, and  $k$  is the release rate as  $\text{CO}_2$  in  $\text{yr}^{-1}$ . Thus stocks in kgC give release in  $\text{kgCO}_2 \text{ yr}^{-1}$ . Repeated disturbances must use the remaining carbon and disjoint affected cohorts. Total forest release is  $F = \sum_h F_h$ : per-tree mass, abundance, exposure and release history all enter. A class containing lighter but numerous affected stems can supply more carbon or annual release than a sparse large-tree class. Partial crown and branch losses, collateral mortality, understory removal and repeated damage belong in degradation accounting even when land remains classified as forest. Regrowth and changes in future uptake are separate terms. These definitions are consistent with the cohort formulation in Section 6 and degradation-process distinctions in Lapola et al. [20].

## 5 Spatial, temporal and accounting boundaries

### 5.1 Stocks, transfers and gas-equivalent quantities

Aboveground biomass is dry biological mass; aboveground carbon is the carbon contained in that mass. Although  $1 \text{ Pg} = 1 \text{ Gt}$ , a stock in  $\text{PgC}$  cannot be inserted into a flow in  $\text{GtC yr}^{-1}$ . For carbon actually emitted as  $\text{CO}_2$ , the conversion is

$$E_{\text{CO}_2} = (44/12)E_C. \quad (8)$$

It preserves, rather than creates, the time dimension. Movement from live wood to dead wood or harvested products is not necessarily an atmospheric emission in the same year. Carbon pools, transfers and eventual release therefore require explicit representation [15].

A multi-gas inventory instead applies a selected climate-equivalence convention,

$$T_{\text{GHG}} = \sum_g \text{GWP}_{100,g} E_g. \quad (9)$$

This is not a conserved carbon-mass balance. A  $\text{CO}_2$  component can legitimately form the numerator of a  $\text{CO}_2\text{e}$  share because the equivalence factor for  $\text{CO}_2$  is one; the denominator must still specify the other gases, time horizon and accounting boundary. Gas conversion does not make a net land-use category equivalent to gross tropical deforestation.

Throughout, a *gross component* means an explicitly positive emission or removal within the selected account. It does not mean that an anthropogenic inventory includes all ecosystem respiration or gross primary production. Those ecosystem fluxes cannot be added indiscriminately to an inventory of human-induced emissions.

## 5.2 The contemporary reference

**Table 4:** Reported reference quantities through 2025. Values are rounded source estimates or projections, not recalibrated forest-carbon results. LULUCF denotes land use, land-use change and forestry.

| Quantity                         | Annual value                           |  | Boundary and status                                                                      |
|----------------------------------|----------------------------------------|--|------------------------------------------------------------------------------------------|
| EDGAR GHG excluding LULUCF       | 54.1 GtCO <sub>2</sub> e               |  | 2025 inventory; AR5 GWP <sub>100</sub>                                                   |
| EDGAR GHG including net LULUCF   | Approximately 55.0 GtCO <sub>2</sub> e |  | Same reported account including its land balance                                         |
| EDGAR Forest Land                | −5.5 GtCO <sub>2</sub>                 |  | Living-biomass net removal; remaining forest and land converted to forest                |
| EDGAR Deforestation              | +3.7 GtCO <sub>2</sub>                 |  | Net category, all regions; includes tropical forest-fire CO <sub>2</sub>                 |
| EDGAR Fires                      | Approximately 2.2 GtCO <sub>2</sub> e  |  | Allocated fire account; tropical forest-fire CO <sub>2</sub> excluded to avoid overlap   |
| EDGAR Organic Soils              | About +1.1 GtCO <sub>2</sub>           |  | Reported CO <sub>2</sub> subtotal; the category also includes N <sub>2</sub> O           |
| EDGAR Other                      | See signed balance                     |  | Non-biomass forest pools and remaining land categories; not independently extracted here |
| EDGAR net LULUCF                 | About +0.9 GtCO <sub>2</sub> e         |  | Sum across the five categories above                                                     |
| GCB net land-use CO <sub>2</sub> | About +4.1 GtCO <sub>2</sub>           |  | 2025 projection, low confidence; different attribution convention                        |

The booklet’s Section 3, Figure 6 and Annex 2 define these entries [21]. Forest Land includes land converted to forest; tropical forest-fire CO<sub>2</sub> belongs to Deforestation, with non-CO<sub>2</sub> fire gases assigned separately. The organic-soil number is a CO<sub>2</sub> subtotal. The rounded closure residual,  $0.9 - [3.7 - 5.5 + 2.2 + 1.1] \approx -0.6 \text{ GtCO}_2\text{e yr}^{-1}$ , consequently absorbs missing subcomponents and rounding as well as Other. The non-land gas composition is 73.8% fossil CO<sub>2</sub>, 17.6% CH<sub>4</sub>, 5.3% N<sub>2</sub>O and 3.3% F-gases; power and heat contribute nearly 30%. The globally implemented Tier 1 Gain/Loss forest account combines land-change products with default growth, expansion, density and allocation factors, including inputs predating the inventory year.

EDGAR’s net +0.9 GtCO<sub>2</sub>e and GCB’s +4.1 GtCO<sub>2</sub> projection refer to different gases, land definitions and attribution rules [21, 22]. A primary harmonization study makes the definitional effect explicit: for 2000–2020, Grassi et al.’s bookkeeping-model +4.8 and national-inventory -1.9 GtCO<sub>2</sub> yr<sup>−1</sup> differ by 6.7; adding the model-estimated managed-forest environmental sink of -6.4 reduces that gap to 0.3 [23]. This demonstrates a consequential reporting difference, separate from the physical calibration of tree mass.

## 5.3 Pantropical coverage, worldwide extension and map dates

The initial map’s extent and epoch are defined in the [status box](#). Baccini’s predictor covers woody vegetation beyond forest-only formations; a global height predictor does not extend the carbon map’s geographical support [16].

Let  $P$  be the actual mapped pantropical domain and  $W \supseteq P$  a harmonized worldwide aboveground-

vegetation domain. For the same carbon pool and reference epoch  $t_0$ , spatial additivity gives

$$C_W^{\text{AG}}(t_0) = C_P^{\text{AG}}(t_0) + C_{W \setminus P}^{\text{AG}}(t_0).$$

Retaining 723.97 PgC for  $C_P^{\text{AG}}(t_0)$ , a complete estimate on this wider support includes additional positive stock outside  $P$ , including the Australian tropics and temperate and boreal regions. Its increment is not quantified by the estimate reported on arXiv. Area-ratio scaling or transfer of one tropical correction factor to all other vegetation would introduce unsupported density and calibration assumptions. For a strictly forest-only target  $F$ , the appropriate partition is instead

$$C_F^{\text{AG}}(t_0) = C_{P \cap F}^{\text{AG}}(t_0) + C_{F \setminus P}^{\text{AG}}(t_0),$$

so the mapped forest contribution must first be distinguished from other included vegetation. Roots, dead organic matter and soil carbon are additional pools, not quantities already represented by the aboveground total.

Table 5 separates predictor acquisition, field calibration and nominal output epoch [1, 16, 24]. Those dates define a multi-date measurement chain; publication alone performs no temporal stock update.

**Table 5:** Spatial and temporal provenance of the pantropical stock estimate. Publication year, predictor acquisition and nominal output epoch have different meanings. The 2025 global emissions account in Table 4 is a separate annual quantity.

| Source / publication                   | Spatial role                                               | Reference period                             | Interpretation                                                             |
|----------------------------------------|------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------|
| Baccini et al. (2012)                  | Pantropical carbon-map predictor; Australia excluded       | Map: 2007–2008; field calibration: 2008–2010 | 2012 dates the publication, not the mapped carbon.                         |
| Simard et al. (2011)                   | Global canopy-height predictor                             | GLAS L3C and tree cover: 2005                | Predictor coverage and acquisition date do not redefine the carbon output. |
| Arellano-P. and Rangel-Ch. (2015/2016) | Calibrated pantropical carbon; Australian tropics excluded | Nominal map epoch: 2007–2008                 | 723.97 PgC estimate reported on arXiv; p. 23 specifies the epoch.          |

Geographic completion and temporal updating are separate operations. On a fixed, consistently tracked pool and mask, moving from  $t_0$  to a later date requires the intervening gains, losses and pool transfers. Extending annual accounts worldwide instead sums the signed regional contributions:  $N_W(t) = \sum_d [E_d(t) - U_d(t)]$  over disjoint regions and aligned attribution rules. Additional regions add positive stock and may add both emissions and removals; their net annual contribution can have either sign. Thus a larger worldwide standing stock does not, by itself, establish a larger 2025 net land flux or a proportional change in greenhouse-gas shares.

## 6 The stock-to-flux transfer identity

### 6.1 Matched support and fixed release operators

Consider spatial or cohort units  $i = 1, \dots, n$  with positive baseline carbon  $C_i$ , local correction factors  $r_i \geq 0$ , and non-negative annual release weights  $w_i$ . The units, carbon pools and disturbance histories are held fixed. The weight has dimensions  $\text{yr}^{-1}$  when a stock is converted to an annual flow. Define

$$\mu_i = \frac{C_i}{\sum_j C_j}, \quad r_C = \sum_i \mu_i r_i, \quad r_F = \frac{\sum_i C_i r_i w_i}{\sum_i C_i w_i}, \quad (10)$$

with  $\mathbb{E}_\mu[w] > 0$ . Carbon-to- $\text{CO}_2$  conversion cancels in this ratio if applied consistently.

**Proposition 6.1** (Stock-to-flux transfer). *Under these conditions,*

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w)}{\mathbb{E}_\mu[w]}. \quad (11)$$

Consequently  $r_F = r_C$  if and only if the stock-weighted covariance is zero. Moreover,

$$\min_i r_i \leq r_F \leq \max_i r_i, \quad |r_F - r_C| \leq \sigma_\mu(r) \text{CV}_\mu(w). \quad (12)$$

*Proof.* Equation (10) gives  $r_F = \mathbb{E}_\mu[rw]/\mathbb{E}_\mu[w]$ . Substitution of  $\mathbb{E}_\mu[rw] = \mathbb{E}_\mu[r]\mathbb{E}_\mu[w] + \text{Cov}_\mu(r, w)$  proves the identity. A positive denominator gives the equivalence. The first bound follows because  $r_F$  is a convex combination of  $r_i$  with weights proportional to  $C_i w_i$ . The second follows from Cauchy–Schwarz and  $\text{CV}_\mu(w) = \sigma_\mu(w)/\mathbb{E}_\mu[w]$ .  $\square$

The result is the covariance decomposition of a weighted mean, algebraically identical to the selection term of the Price equation [25]. Its scientific role is to identify the missing empirical quantity: the association between where carbon is misestimated and where that carbon is released. Statistical independence is sufficient but not necessary; zero covariance under the specified stock weights is the exact condition. A constant  $r_i$  is another sufficient condition. Neither condition follows from a ratio of published global totals on different masks.

**Table 6:** Two-unit counterexample. Baseline stocks are (50, 50) and corrections (1, 3) in both rows, so  $r_C = 2$ . Only the annual release location changes. Units are synthetic.

| Annual release weights | $\mathbb{E}_\mu[w]$ | $\text{Cov}_\mu(r, w)$ | $r_C$ | $r_F$ |
|------------------------|---------------------|------------------------|-------|-------|
| (0.02, 0)              | 0.01                | −0.01                  | 2     | 1     |
| (0, 0.02)              | 0.01                | +0.01                  | 2     | 3     |

## 6.2 Census changes, offsets and state-dependent calibration

For a stable scalar correction, two-census stock change obeys  $\Delta C^* = r\Delta C$ . A time-varying calibration instead gives the exact decomposition

$$C_1^* - C_0^* = r_0(C_1 - C_0) + C_1(r_1 - r_0). \quad (13)$$

The second term must be interpreted rather than automatically labelled instrument drift. It can reflect a changed calibration procedure, but it can also arise from a fixed size- or age-dependent calibration evaluated at different biological states. A fixed additive offset cancels for an unchanged set of units; recruitment, mortality or support changes can prevent that cancellation. A stable multiplicative error generally changes stock differences. A stand-wide scalar therefore requires justification for heterogeneous trees and changing composition.

This matters for repeated-census forest sinks. Plot-network analyses estimate biomass dynamics and net live-biomass accumulation using repeated tree observations and biomass relationships [26]. A shared, stable multiplicative mass correction changes these inferred gains and losses, not only disturbance emissions. However, a net increase in live aboveground biomass is not gross primary production or automatically net ecosystem–atmosphere uptake. Mortality transfers mass into other pools; decomposition, roots, soils, respiration and exports are needed to close the atmospheric boundary [15]. The paired formulation below applies only after those quantities have been aligned.

### 6.3 Disturbance cohorts and carbon fate

For area  $A_{ia\tau}$  disturbed by process  $a$  in year  $\tau$ , let  $h$  index disjoint tree-size, organ and fate compartments, with pre-disturbance carbon density  $c_{iah\tau}$  and affected fraction  $d_{iah\tau}$ . Set  $L_{iah\tau} = A_{ia\tau}c_{iah\tau}d_{iah\tau}$ . With  $k_{iah}^{\text{CO}_2}(t - \tau)$  the annual release rate ( $\text{yr}^{-1}$ ) for  $\text{CO}_2$  in year  $t$ ,

$$F_{\text{CO}_2}(t) = \frac{44}{12} 10^{-9} \sum_{i,a,h,\tau \leq t} L_{iah\tau} k_{iah}^{\text{CO}_2}(t - \tau). \quad (14)$$

Here  $L$  is in  $\text{MgC}$  and the output is  $\text{GtCO}_2 \text{ yr}^{-1}$ . For annual bins,  $k\Delta t$  is the released fraction with  $\Delta t = 1$  yr; cumulative fate fractions must not exceed the available cohort carbon. Other fate fractions represent retained live biomass, dead material, products, charcoal, exports or carbon released in other gases. Their mass bookkeeping must conserve carbon over the specified horizon. Regrowth is a separate uptake term and cannot be represented by making a gross release kernel negative.

The fixed-operator identity is exact when the correction changes carbon amounts but not areas, disturbance fractions or fate kernels. When those inputs also change,  $r_F$  must be recomputed from the revised products rather than attributed exclusively to  $r_C$ . An index can include cohort and pool, so the identity is compatible with delayed release; it does not remove the need to measure those delays.

### 6.4 Size-dependent release and the timing of a stock correction

The opposite-order sign used below is Chebyshev's sum inequality for similarly or oppositely ordered sequences, here written under stock weights. Its application concerns physical release from a defined cohort, rather than a rule for booking inventory stock losses. Combustion measurements in a rainforest clearing distinguish markedly different completeness among fuel-size components [27]. This supplies a physical candidate for the covariance in Proposition 6.1, not a universal monotone law: diameter, moisture, decay, burn history and exposure must be controlled. Where structural underestimation is concentrated in large wood and those compartments burn less completely, their carbon correction receives less weight in initial emissions. Extraction of boles into retained products can reinforce the initial delay, provided products and exports are tracked rather than lost from the boundary [15].

**Proposition 6.2** (Opposite ordering and cumulative release). *On a fixed affected-carbon cohort with weights  $\mu_i = C_i/C_\Sigma$ , let  $p_{it} \geq 0$  be the fraction of compartment  $i$  emitted as  $\text{CO}_2$  in annual bin  $t$ , and  $P_i(H) = \sum_{t \leq H} p_{it}$ . If  $(r_i - r_j)(p_{i0} - p_{j0}) \leq 0$  for every pair, then, for positive initial release,  $r_{F,0} \leq r_C$ . For a horizon with positive cumulative release,*

$$r_{F,\leq H} = r_C + \frac{\text{Cov}_\mu(r, P(H))}{\mathbb{E}_\mu[P(H)]}. \quad (15)$$

*If  $P_i(\infty) = 1$  for every compartment,  $r_{F,\leq H} \rightarrow r_C$ . The same conclusion holds for any common positive limiting fraction.*

*Proof.* The covariance identity

$$\text{Cov}_\mu(r, p_0) = \frac{1}{2} \sum_{i,j} \mu_i \mu_j (r_i - r_j)(p_{i0} - p_{j0}) \quad (16)$$

has non-positive summands under the ordering assumption. Apply Proposition 6.1 to  $p_0$  and to  $P(H)$ . A constant positive limiting release fraction has zero covariance. Strict initial inequality holds if some positive-weight pair has a strictly negative product.  $\square$

The stock reference here is *carbon affected in the matched cohort*, not all standing pantropical carbon. The premise is a joint ordering to be tested, not a consequence of tree size alone; diameter-dependent drying

or repeated burning can alter it. Complete oxidation as  $\text{CO}_2$  within the tracked boundary is stronger than clearing a stand. Persistent charcoal, retained products, export outside that boundary and other gaseous carbon can leave unequal limiting fractions and nonzero cumulative covariance. A delayed correction need not reappear in every subsequent year. Under complete accounting, its cumulative allocation does recover the stock multiplier.

A synthetic two-compartment example uses stocks (50, 50), corrections (1, 3) and first-year fractions (0.8, 0.1). Then  $r_C = 2$  but  $r_{F,0} = 55/45 = 1.222\dots$ . If the remaining fractions (0.2, 0.9) are subsequently oxidized, their combined later multiplier is  $145/55 = 2.636\dots$ , and the lifetime multiplier is  $200/100 = 2$ . These synthetic transfers describe a physical release history. Applying them to EDGAR requires identifying whether the target is conversion-year stock-loss booking or atmospheric release: the inclusion of tropical fire  $\text{CO}_2$  in Deforestation does not itself make a decay kernel part of the inventory operator. Section 6.5 distinguishes those accounting conventions.

### 6.5 Annual mixtures, disturbance trends and inventory conventions

A year- $t$  release account combines cohorts of different ages. Assume fixed compartment shares  $C_i > 0$  per unit disturbance, fixed corrections  $r_i$ , annual fate fractions  $p_{i,s} \geq 0$ , and a non-negative disturbance history  $D(t-s)$ . For annual time bins, let

$$W_i(t) = \sum_{s \geq 0} D(t-s)p_{i,s}, \quad r_{F,t} = \frac{\sum_i C_i r_i W_i(t)}{\sum_i C_i W_i(t)} = r_C + \frac{\text{Cov}_\mu(r, W(t))}{\mathbb{E}_\mu[W(t)]}. \quad (17)$$

Here  $C_i D$  is affected carbon, and division by the one-year bin width gives an annual rate. The common stock weights presume stable compartment composition; changing forest types or disturbance selection require a cohort-specific extension.

**Corollary 6.3** (Stationary mixtures and a two-lag trend criterion). *If  $D$  is constant over all contributing ages and  $\sum_s p_{i,s} = 1$  for every compartment, then  $r_{F,t} = r_C$ . With two lags,  $p_{i,1} = 1 - p_{i,0}$ , and  $D_0 = D(t)$ ,  $D_1 = D(t-1)$ ,*

$$r_{F,t} - r_C = \frac{(D_0 - D_1) \text{Cov}_\mu(r, p_0)}{D_1 + (D_0 - D_1) \mathbb{E}_\mu[p_0]}. \quad (18)$$

*The denominator is assumed positive. Oppositely ordered corrections and initial releases therefore suppress the annual multiplier during increasing disturbance, enhance it during decreasing disturbance, and cancel under stationarity.*

*Proof.* Stationarity and complete ultimate release make each  $W_i$  equal to the same constant. For two lags,  $W_i = D_1 + (D_0 - D_1)p_{i,0}$ . Substitution into (17) proves the formula and sign.  $\square$

For  $C = (50, 50)$ ,  $r = (1, 3)$  and  $p_0 = (0.8, 0.1)$ , stationary clearing gives multiplier 2; doubling the latest cohort gives  $255/145 = 51/29$ , and halving it gives  $172.5/77.5 = 69/31$ . The comparison concerns the multiplier relative to  $r_C$ , while corrected release remains larger than the baseline. Longer histories use the full convolution. Under conversion-year stock-loss booking, weights instead describe affected area and carbon; EDGAR's Tier 1 description does not specify a size-resolved decay kernel. The physical release operator is needed for atmospheric reconciliation, independently of that inventory convention [21].

Edge-associated biomass contrasts provide an empirical reason to measure the covariance rather than assume it vanishes. Chaplin-Kramer et al. report approximately 25% lower biomass within 500 m of tropical forest edges and estimate nearly 10% overstatement by Tier 1 carbon-stock methods that omit these effects [28]. These findings establish a spatial carbon contrast. If a uniform reference overstates edge stocks and independently observed clearing weights concentrate at those edges, the correction–disturbance covariance is negative. The biomass study alone does not measure that joint clearing distribution or establish a universal global sign.

## 7 Paired source–sink corrections

### 7.1 General reconciliation first; common-stock transfer as a special case

Let  $F, U > 0$  be a specified source and positive removal magnitude, with corrected quantities supplied by appropriate, potentially different measurement operators. Define  $r_F = F^*/F$ ,  $r_U = U^*/U$  and the signed account  $N = F - U + Z$ .

**Proposition 7.1** (Operator-independent net correction). *For any such corrected components,*

$$\boxed{\Delta N = (r_F - 1)F - (r_U - 1)U + \Delta Z.} \quad (19)$$

*No common standing-stock basis or size-independent calibration is required.*

*Proof.* Subtract  $F - U + Z$  from  $F^* - U^* + Z^*$  and substitute the two component ratios.  $\square$

A more restrictive representation sets

$$F = \sum_i C_i w_i^+, \quad U = \sum_i C_i w_i^-, \quad w_i^\pm \geq 0, \quad (20)$$

with positive stocks, fixed weights and the same stock correction  $r_i$  multiplying both terms *within each unit*. Applying [Proposition 6.1](#) separately gives

$$r_F = r_C + \frac{\text{Cov}_\mu(r, w^+)}{\mathbb{E}_\mu[w^+]}, \quad r_U = r_C + \frac{\text{Cov}_\mu(r, w^-)}{\mathbb{E}_\mu[w^-]}. \quad (21)$$

Consequently, only in that shared-stock fixed-operator case,

$$\Delta N = (r_C - 1)(F - U) + C_\Sigma \text{Cov}_\mu(r, w^+ - w^-) + \Delta Z. \quad (22)$$

The proof is substitution in (19), using  $F/\mathbb{E}_\mu[w^+] = U/\mathbb{E}_\mu[w^-] = C_\Sigma$ . Spatially heterogeneous  $r_i$  remain permissible, but size-dependent change *within* a growing unit cannot generally be replaced by that unit's stock-level multiplier. Positive regrowth from zero stock is outside this representation, and intact-plot increments require endpoint calibration when  $r$  changes with state. Thus (19), not (22), governs the general application. Carbon units, atmospheric boundaries and overlap must be reconciled before either is used; a common CO<sub>2</sub> conversion can then be applied.

### 7.2 Regrowth from zero stock requires an increment basis

Secondary-forest recovery is a trajectory through age, composition and environment, not a fixed fraction of pre-clearing carbon. Neotropical observations document age-dependent and site-dependent biomass recovery after forest removal [29]. Accordingly, let cohort  $a$  occupy fixed area  $A_a$ , with baseline live-carbon densities  $c_{a0}, c_{a1}$  at two ages separated by  $\Delta t > 0$ . Let  $c_{ak}^* = r_{ak} c_{ak}$  denote a specified endpoint recalibration. Areas in hectares and densities in MgC ha<sup>-1</sup> give gains in MgC yr<sup>-1</sup> when the interval is in years. For a selected non-decreasing baseline trajectory, define  $g_a = (c_{a1} - c_{a0})/\Delta t \geq 0$ . Its corrected increment satisfies

$$g_a^* = \frac{r_{a1}c_{a1} - r_{a0}c_{a0}}{\Delta t} = r_{a0}g_a + \frac{c_{a1}(r_{a1} - r_{a0})}{\Delta t}. \quad (23)$$

This is an age-cohort application of the endpoint identity, not evidence of temporal calibration drift. In particular,  $c_{a0} = 0 < c_{a1}$  gives  $g_a^* = r_{a1}g_a$ : the cohort is retained and its initially undefined stock ratio need not be estimated.

**Proposition 7.2** (Increment-weighted regrowth correction). *For the fixed cohort areas, let  $I_+ = \{a : g_a > 0\}$ ,  $I_0 = \{a : g_a = 0\}$ , and  $K = \sum_a A_a g_a > 0$ . Define  $q_a = g_a^*/g_a$  on  $I_+$  and  $v_a = A_a g_a^*/K$ . Then*

$$K^* = \sum_{a \in I_+} A_a q_a g_a + K_0^*, \quad \frac{K^*}{K} = \mathbb{E}_v[q] + \frac{K_0^*}{K}, \quad K_0^* = \sum_{a \in I_0} A_a g_a^*. \quad (24)$$

*Proof.* Partition the sum  $K^* = \sum_a A_a g_a^*$  into  $I_+$  and  $I_0$ , and substitute the definition of  $q_a$  on the former. The weights  $v_a$  sum to one because zero increments contribute nothing to  $K$ .  $\square$

The averaging weights are *baseline gains*, not initial stocks. If  $K_0^* = 0$ , the replacement factor is a gain-weighted mean; a uniform endpoint correction supplies the familiar common-scale special case. A cohort with zero baseline increment but nonzero revised increment contributes additively and cannot be discarded. If  $K = 0$ , a total multiplier is undefined even when  $K^*$  is positive. A newly represented pool with zero baseline carbon at both endpoints also needs an additive model rather than a finite multiplicative correction. Baseline losses and corrected sign reversals require explicitly signed accounting instead of calling every term a removal.

For example, a zero-stock hectare that gains 5 MgC over one year gains 6 after a final-stock correction of 1.2; it remains part of the analysis despite having zero initial carbon. These are illustrative numbers.  $K$  is live-carbon accumulation over the selected interval, not gross photosynthesis or a complete atmospheric removal. Regrowth cohorts must be disjoint from any intact-forest accumulation subtotal, and mortality transfers, dead pools, soil exchange and exported carbon must be reconciled before  $K$  contributes to an atmospheric account.

### 7.3 Common-error cancellation and its limits

For a uniform stable multiplier  $r$ , fixed  $Z$  gives

$$\Delta N = (r - 1)(F - U), \quad \eta = \frac{|F - U|}{F + U}, \quad \frac{|\Delta N|}{|\Delta F| + |\Delta U|} = \eta \quad (r \neq 1). \quad (25)$$

$\eta$  lies between zero and one. Thus a nearly balanced pair can undergo a large revision of both components with a small *absolute* adjustment of their net difference. The relative change of the nonzero calibrated net difference is still  $r - 1$ ; cancellation must not be described as generally small relative net bias. Exact balance leaves the net unchanged, regardless of the common multiplier.

The converse sensitivity is just as important. Write  $r_F = r + \delta/2$  and  $r_U = r - \delta/2$ . Then

$$N^* = r(F - U) + \frac{\delta}{2}(F + U) + Z^*. \quad (26)$$

A small differential correction is multiplied by the gross sum, not the net difference. Near balance, unequal source and sink corrections can dominate the common-scale contribution and reverse the sign of net exchange. Pairing therefore creates a testable mechanism for hidden gross bias, not an automatic means of making any proposed stock compatible with atmospheric constraints.

A source-reported process pair provides an empirical scale. For 2015–2024, Global Carbon Budget 2025 reports deforestation emissions of approximately 1.9 GtC yr<sup>-1</sup> and forest (re-)growth removals of 1.3 GtC yr<sup>-1</sup> [22, Section 3.2.2]. As a selected pair, these yield  $F - U = 0.6$ ,  $F + U = 3.2$  and  $\eta = 0.1875 \simeq 0.19$ ; an illustrative  $\delta = 0.1$  contributes 0.16 GtC yr<sup>-1</sup> through (26). It is not the complete net land-use account.

The primary-source decomposition separates permanent deforestation (1.1) and re/afforestation uptake (0.6) from shifting-cultivation emissions and regrowth; subtraction from the rounded totals gives 0.8 and 0.7 GtC yr<sup>-1</sup> for the latter pair. Table 7 keeps the process definitions explicit. The shifting-cultivation

**Table 7:** Selected GCB land-use process pairs for 2015–2024 [22, Section 3.2.2].  $U$  is a positive removal magnitude; rates are in  $\text{GtC yr}^{-1}$ . Shifting-cultivation values are derived from rounded aggregates. The final column is the synthetic sensitivity  $0.1(F + U)/2$ , not an observed calibration correction.

| Pair                 | $F$ | $U$ | $F - U$ | $\eta$ | Shift at $\delta = 0.1$ |
|----------------------|-----|-----|---------|--------|-------------------------|
| Selected total       | 1.9 | 1.3 | 0.6     | 0.19   | 0.160                   |
| Permanent change     | 1.1 | 0.6 | 0.5     | 0.29   | 0.085                   |
| Shifting cultivation | 0.8 | 0.7 | 0.1     | 0.07   | 0.075                   |

entries are differences of rounded source estimates; their uncertainties cannot be derived by subtracting marginal model ranges.

The two-lag and finite-increment results explain why shared land does not imply identical annual correction factors. A shifting-cultivation cycle can share a carbon calibration across clearing and regrowth, but vegetation age, organ allocation and lagged cohorts can still produce  $r_F \neq r_U$ . Permanent loss and re/afforestation need not involve the same regions or populations. GCB identifies China, the USA and the EU27 as leading re/afforestation removal contributors, with  $0.3 \text{ GtC yr}^{-1}$  combined—about half of the rounded permanent-removal total [22, Executive summary and Section 3.2.2]. Country boundaries do not identify tropical or extratropical pixels; a tropical-only recalibration cannot be assigned uniformly to that global sink.

Let  $p_F$  and  $p_U$  be the fractions of the selected source and removal affected by a local common multiplier  $r$ , leaving their complements unchanged. Then the specified scenario gives

$$\Delta N = (r - 1)(p_F F - p_U U),$$

where the fractions are *flux-weighted exposure*, not simply shared land area. Unequal exposure produces the differential case even for one locally shared multiplier. For example,  $r = 1.2$  on the permanent pair with  $p_F = 1$ ,  $p_U = 0$  changes the net by 0.22, compared with 0.10 when both are fully exposed. These are testing scenarios, not inferred geographic fractions. Process- and region-resolved carbon calibration determines which scenario is admissible; neither cancellation nor source-only amplification is imposed on the actual inventory.

The EDGAR entries in Table 4 are signed reporting categories with a different boundary; their opposite signs do not make them the same gross process pair. The GCB decomposition is used here specifically to separate the above mechanisms.

#### 7.4 Size-dependent calibration and a directional census test

Size- and architecture-dependent discrepancies motivate a differential-correction test across all represented stem classes [1, 19, 30]. Large-tree studies provide one calibration domain, while the intermediate-stem contributions in Section 4.4 require their own event weights. Neither individual tree size nor a selected validation sample establishes the dominant contribution to population-level release, or that mortality receives a larger correction than growth. To see the distinction, let  $x$  be a tree’s baseline carbon and let a stable, continuously differentiable, zero-intercept calibration be  $m(x) = r(x)x$ . A surviving tree growing from  $x_0$  to  $x_1 > x_0$  has gain multiplier

$$q_G(x_0, x_1) = \frac{m(x_1) - m(x_0)}{x_1 - x_0}, \quad m'(x) = r(x) + xr'(x). \quad (27)$$

The secant equals the average of  $m'$  over that interval. Mortality of a tree with assigned pre-death stock  $x_d$ , in contrast, contributes  $m(x_d)/x_d = r(x_d)$ . An increasing stock multiplier is not the same function as the growth multiplier. Species, architecture and changing organ allocation can require a multivariate calibration; the scalar construction isolates the logical distinction without imposing a universal fitted curve.

Let  $G, L > 0$  denote positive gains and losses in a matched live-carbon census account, with net accumulation  $S = G - L$ . Sum calibrated survivor increments and consistently accounted recruitment into  $G^*$ ; use matched pre-death stocks and other specified live-pool losses for  $L^*$ . The resulting ratios  $r_G = G^*/G$  and  $r_L = L^*/L$  give

$$S^* = r_G G - r_L L = r_G S - (r_L - r_G)L, \quad S^* - S = (r_G - 1)G - (r_L - 1)L. \quad (28)$$

Recruitment across an inventory threshold is an entry into the observed pool, not necessarily carbon assimilated wholly during that interval. Diameter thresholds, survivor shrinkage, missing stems and death-time conventions must therefore be fixed or reconciled. Mortality here is loss from the live pool; atmospheric release follows its carbon-fate operator, not immediate emission by definition.

**Directional hypothesis  $H_\Delta$ .** In a specified, independently calibrated census population, the event-weighted loss correction exceeds the gain correction by a preregistered meaningful margin,  $r_L - r_G > \epsilon_\Delta \geq 0$ . This predicts  $S^* < r_G S$ , lower net live-carbon accumulation than under common gain scaling. For  $S^*$  to be smaller than the *uncorrected* sink, the exact condition is  $(r_L - 1)L > (r_G - 1)G$ . When  $S > 0$  and  $r_G \geq 1$ , this condition implies the zero-margin directional condition  $r_L > r_G$ ; it need not imply an independently chosen positive margin  $\epsilon_\Delta$ . Without those conditions, calling it a universally stronger inequality is unwarranted. Neither prediction follows solely from larger average size among dead trees. The hypothesis concerns measured event-weighted corrections. A prespecified interval for their difference wholly above the margin supports the directional claim; an interval wholly at or below the margin opposes it, while overlap leaves the test unresolved at that precision.

**Table 8:** Synthetic differential-census cases with  $G = 120$ ,  $L = 100$  and  $S = 20$  in arbitrary carbon units over a common interval. The second row uses one increasing size-calibration function, not fitted forest observations.

| Case                           | $r_G$ | $r_L$ | $G^*$ | $L^*$ | $S^*$ |
|--------------------------------|-------|-------|-------|-------|-------|
| Loss-weighted hypothesis       | 1.20  | 1.50  | 144.0 | 150.0 | -6.0  |
| Increasing-size counterexample | 1.85  | 1.50  | 222.0 | 150.0 | 72.0  |

The counterexample in Table 8 uses  $m(x) = x(1 + 0.005x)$  in the specified synthetic units: 12 survivors grow from 80 to 90, while one larger tree of stock 100 dies. Then  $r_G = 1.85 > r_L = 1.50$  and  $S^* = 72$ , despite the strictly increasing  $r(x)$  and the larger dead tree. Underestimation of large stock levels therefore does not determine the corrected sink's direction. This counterexample does not deny differential amplification; it identifies the finite-increment evidence needed to turn that mechanism into a defensible prediction.

## 7.5 Operational census screens

For a common calibration applied to a matched census, let  $\bar{x}_L$  be the mortality-stock-weighted mean tree carbon and  $\bar{x}_G$  the gain-weighted mean of survivor interval midpoints. The quadratic family  $m(x) = ax + kx^2$ , with  $k > 0$ , corrects losses more than gains exactly when  $\bar{x}_L > 2\bar{x}_G$ . A specified positive effect margin additionally requires the fitted magnitude of  $k$ .

For  $m(x) = ax^b$ ,  $b > 1$ , the exact finite-interval screen instead uses the ratio of loss- and interval-gain-weighted  $(b - 1)$ -power means, which must exceed  $b^{1/(b-1)}$ . An arithmetic ratio below two excludes the quadratic direction but does not exclude every power exponent. Near  $b = 1$ , the log-moment difference determines the first-order direction and the differential tends to zero. Supplementary Note S2 gives the definitions, proofs, threshold behaviour and counterexample. These screens use baseline event distributions to identify admissible directions; independent calibration and uncertainty remain necessary to estimate their size.

## 7.6 Gross-source and net-land shares are different quantities

For a simplified account with other positive sources  $O$  and  $Z = 0$ , define

$$s_F^{\text{gross}} = \frac{F}{O + F}, \quad s_N^{\text{net}} = \frac{F - U}{O + F - U}. \quad (29)$$

The first is the forest fraction of the selected positive source sum. The second is a signed net-land contribution relative to a net total; it can be negative and becomes unstable as that total approaches zero.  $F/(O + F - U)$  is a third ratio: a gross component divided by a net total. It is not a conventional composition of non-negative shares and can exceed one. When  $Z$  contains additional sources and removals, those components must be explicitly partitioned before forming a gross-source sum.

**Table 9:** Synthetic paired-correction example:  $F = 6$ ,  $U = 5.8$ ,  $O = 50$ ,  $Z = 0$ . Flow units are arbitrary common annual mass units, not 2025 observations. The last two columns follow (29); percentages have different denominators.

| Case                | $r_F$ | $r_U$ | $F^*$ | $U^*$ | $F^* - U^*$ | Gross share (%) | Net share (%) |
|---------------------|-------|-------|-------|-------|-------------|-----------------|---------------|
| Reference           | 1.0   | 1.0   | 6.00  | 5.80  | 0.20        | 10.71           | 0.40          |
| Source only         | 2.5   | 1.0   | 15.00 | 5.80  | 9.20        | 23.08           | 15.54         |
| Paired uniform      | 2.5   | 2.5   | 15.00 | 14.50 | 0.50        | 23.08           | 0.99          |
| Paired differential | 2.5   | 2.6   | 15.00 | 15.08 | -0.08       | 23.08           | -0.16         |

In the uniform example, corrections of +9 and +8.7 in gross emission and removal magnitudes change the net by only +0.3. The cancellation ratio is about 0.01695. With a slightly larger sink multiplier, the revised net becomes negative. These are algebraic demonstrations of identifiability and denominator dependence, not inferred global forest fluxes. They motivate independent constraints on source and removal weights, rather than a preferred multiplier.

## 7.7 Independent measurements remain decisive

A common allometric calibration can affect census-derived quantities, but it need not affect an independently measured atmospheric concentration increment, fossil-fuel statistic or tracer constraint in the same way. The theoretical cancellation concerns shared calibration of specified components; regrowth and live-census accumulation enter only after their increment operators and pool boundaries have been aligned. It cannot be imposed on independently constrained terms merely to force closure.

Likewise, living-biomass uptake, managed-land net removals and bookkeeping net land-use emissions are not interchangeable sink variables. A correction must follow the underlying mass estimator into its valid pool and process, account for time lags, and identify overlap with other entries. Only then can the proposed source–sink revision be tested against a carbon-mass budget. The fact that a budget can contain compensating errors supplies no evidence for a particular compensation [22]. Without a supported offset, an upward revision of actual net forest release increases the corresponding reconstructed atmospheric accumulation, as made explicit in Section 8.1.

## 8 Atmospheric consequences and source attribution

### 8.1 Omitted net sources and the observed carbon balance

For atmospheric carbon mass  $M_{\text{atm}}$ , a harmonized account obeys

$$\frac{dM_{\text{atm}}}{dt} = E_{\text{DD}} + E_{\text{other}} - U_{\text{land}} - U_{\text{ocean}}.$$

Here  $E_{DD}$  is the defined deforestation/degradation release and  $U$  denotes positive removal magnitudes. For corrected-minus-reference differences with the same initial mass,

$$\delta M_{\text{atm}}(t_1) = \int_{t_0}^{t_1} [\delta E_{DD} + \delta E_{\text{other}} - \delta U_{\text{land}} - \delta U_{\text{ocean}}] dt.$$

A missing positive net source places actual accumulation above the account's reconstruction unless other terms offset it. Delayed decomposition assigns release to later years; reduced uptake enters once through the relevant removal term. Direct atmospheric observations may already contain carbon absent from a bottom-up inventory, in which case the correction changes its source attribution rather than the observed concentration [22, 31].

## 8.2 Accounting alternatives and their empirical standing

For the signed account  $T = F + O - U$ , the source, other-source and removal corrections satisfy

$$\Delta T = \Delta F + \Delta O - \Delta U. \quad (30)$$

The three outcomes in Figure 2 retain the substantive inference of forest-carbon recalibration: an independently supported upward correction to annual deforestation and degradation can increase the estimated total, increase the contribution assigned to forest loss within a retained total, or change several sources and removals together. They are conditional cases of one balance, rather than competing physical theories or three measured present-day allocations.

**I. Larger total with other components retained.** For  $\Delta F > 0$  and  $\Delta O = \Delta U = 0$ , the total rises by  $\Delta T = \Delta F$ . On a physical carbon-mass boundary this yields greater reconstructed atmospheric accumulation over the corresponding release history. Delay redistributes release among years; it does not remove the later contribution. This case tests an omitted source, without assuming a compensating sink.

**II. Greater forest attribution within a retained total.** If the total and removals are independently retained,  $\Delta T = \Delta U = 0$  implies  $\Delta O = -\Delta F$ . The forest contribution increases and the aggregate assigned to other sources decreases. Evidence must identify the affected other components; the identity does not assign the adjustment to fossil energy by default. A leading forest-loss contribution is a conditional outcome, requiring  $F^* > \max_j O_j^*$  for positive source categories on a common boundary. A majority requires the stronger condition  $F^* > \sum_j O_j^*$ . An increase in share does not itself establish either ranking.

**III. Joint source–removal revision.** Carbon calibration can affect both release and accumulation, and multiple terms may require reassessment. Their supported corrections enter the same signed identity. The net adjustment can be positive, negative or small; shared calibration does not establish equal multipliers or an automatic cancellation. This case accommodates the measurement dependence developed in Section 7 without prescribing its empirical result.

**Largest-sector inference with the other components retained.** A change in total and a change in sector ranking can occur together under case I. Let the consistently defined land category be  $L = F + Z_L$ , with its other net contribution  $Z_L$  retained, and let  $O_j$  be unchanged peer sectors. Then  $L^* = r_F F + Z_L$  and, for a common positive revised total,

$$L^* > \max_j O_j \iff \frac{L^*}{T^*} > \max_j \frac{O_j}{T^*}.$$

Thus the corrected land category *is the largest sector* whenever its corrected absolute contribution exceeds every retained peer contribution. Renormalizing the percentages cannot undo that ranking. Keeping the other components means keeping their absolute amounts and accounting boundaries, not keeping the

total or their old percentages simultaneously. The sufficient and necessary threshold for positive  $F$  is  $r_F > (\max_j O_j - Z_L)/F$ .

Figure 2 gives a conditional numerical instance on Ecofys's 2010 grouping, alongside the questioned total and double-headed arrow. The source's land-use group combines the 10.3% deforestation/afforestation and land-use CO<sub>2</sub> component with 4.4% agricultural soils; its industry subflows sum to 29.1%, the largest comparison sector [5]. Imposing the historical ratios 723.97/285.1 and 723.97/228.7 on the 10.3% component, while retaining all other absolute contributions, gives land-use-group shares of 26.37% and 30.26% of their respective revised totals. The retained industry shares become 25.12% and 23.79%. The land-use group therefore ranks first at both endpoints in this reconstruction, without reducing industry's absolute emissions. This historical grouping includes agricultural soils and differs from modern EDGAR LULUCF; the component alone exceeds industry only at the upper endpoint. The empirical stock-to-annual transfer and source compatibility remain to be established, as specified in Supplementary Note S3. Neither a present-day LULUCF ranking nor dominance over the combined energy supersector follows from this historical illustration.

The reference labels on the non-land branches of Figure 2 identify baseline contributions, not exact or unchangeable values. Its highlighted outcomes concern the propagation of measurement corrections; graphical prominence is not a quantitative ranking. Gross forest release, net land-use exchange and multi-gas sector accounts are different comparisons and require their own source definitions.

For a specifically defined 2024 two-source CO<sub>2</sub> account, GCB gives fossil emissions 10.3 and net land-use emissions 1.3 GtC yr<sup>-1</sup> [22]. Reassigning  $x$  from fossil to net land use would reverse their ranking only if

$$1.3 + x > 10.3 - x, \quad x > 4.5 \text{ GtC yr}^{-1}.$$

That exceeds nine times the quoted 0.5 GtC yr<sup>-1</sup> fossil standard-uncertainty scale and implies a fossil reduction of approximately 44%. This particular fossil-to-net-land ranking reversal is unsupported by the cited evidence and cannot be inferred from a stock ratio. The calculation does not reject every fixed-total correction: a smaller reattribution, a correction within specified land components, or a comparison of gross forest release with disaggregated sectors is a different claim requiring its own observations. In particular, gross release accompanied by removals cannot be substituted for the net land-use term in this two-source benchmark. A quoted standard uncertainty is not a hard physical bound or a Gaussian tail probability.

### 8.3 Persistent corrections and joint constraints

Define the carbon-budget imbalance  $B = E_{\text{fossil}} + F - U - S_{\text{ocean}} - G_{\text{atm}}$ . Its change is

$$\Delta B = \Delta E_{\text{fossil}} + \Delta F - \Delta U - \Delta S_{\text{ocean}} - \Delta G_{\text{atm}}.$$

The baseline residual, period and joint uncertainty determine admissibility. Selected source constraints are in Table 10. GCB describes the 1959–2024 budget as approximately balanced overall, while reporting a cumulative residual of about +35 GtC, roughly 7% of total emissions; the 2015–2024 mean is below 0.1 GtC yr<sup>-1</sup>. Its –1.7 GtC residual in 2024 instead suggests overestimated total uptake or atmospheric growth that year [22, Sections 3.5 and 3.9]. These temporal constraints are distinct. An uncompensated +0.5 GtC yr<sup>-1</sup> source over 66 years adds 33 GtC to the positive cumulative residual, worsening it to about +68 GtC. Offsets of –33 and –68 GtC would, respectively, preserve the baseline residual and close that constructed account to zero; neither is established by this sensitivity calculation.

These constraints limit particular magnitudes and histories, rather than eliminate the three outcomes in Figure 2. The illustrated large persistent source-only addition worsens the retained positive cumulative residual; the illustrated several-GtC fossil-to-net-land reassignment lacks independent support. Other

**Table 10:** Selected Global Carbon Budget 2025 constraints used to assess source-account alternatives [22]. Annual rates are in  $\text{GtC yr}^{-1}$ . Quoted uncertainties are source estimates, not admissibility ceilings. Annual and decadal estimates are kept distinct.

| Quantity                           | Period    | Reported estimate / implication                                             |
|------------------------------------|-----------|-----------------------------------------------------------------------------|
| Fossil emissions                   | 2024      | $10.3 \pm 0.5$ ; a large downward correction requires independent evidence. |
| Atmospheric growth, surface record | 2024      | $7.9 \pm 0.2$ ; use the detailed atmospheric section.                       |
| Atmospheric growth, decadal mean   | 2015–2024 | $5.6 \pm 0.02$ ; not the uncertainty of a single year.                      |
| Net land-use emissions             | 2015–2024 | $1.4 \pm 0.7$ ; distinct from gross deforestation.                          |
| Environmentally driven land sink   | 2015–2024 | $2.4 \pm 0.8$ ; different attribution from land-use regrowth.               |

periods, spatial domains and component corrections must be evaluated against their corresponding evidence. Joint land-process reconciliation is appropriate when biological calibration affects several components, but is not an assumed resolution of an upward forest-source correction. It still requires independent source, removal and regional constraints; it is not guaranteed compensation. Common scaling of a nearly balanced pair gives a small absolute net change, whereas the differential term  $\delta(F + U)/2$  can produce a large net change or sign reversal. The constraint is the signed multiyear sum and its uncertainty, not a rule that only small-net paired revisions are physically possible.

## 9 A calibration-first programme for forest-carbon accounting

### 9.1 Independent material comparisons

Compare allometric and reconstructed carbon with a common, independently checked reference on matched trees, plots, pools and dates. Retain organ geometry, basic density, tissue fractions, nonmerchantable material and a distinct dead-pool account. For reference  $C_{\text{ref}}$ , report residuals  $e_A = \hat{C}_A - C_{\text{ref}}$  and  $e_R = \hat{C}_R - C_{\text{ref}}$  with their joint uncertainty and dependence on architecture, composition and disturbance. A negative allometric residual associated with missing architectural predictors supports the underestimation mechanism; each pathway’s residual remains subject to the same decision rule.

The class-5 application has neither a recovered independent geometric residual nor a validated joint physical envelope. The Córdoba aggregates lack sound organ volumes and matched map values. Supplementary Note S1 identifies the structural combinations needed to attain the mean and the conditions under which an independently measured upper envelope would reject it. A prespecified equivalence margin permits agreement, positive or negative discrepancy, or an unresolved comparison. Representative area weighting, held-out sites and material-property uncertainty distinguish empirical evaluation from numerical closure of the reconstruction identity.

### 9.2 Calibration, spatial transfer and disturbance

Build vegetation-specific reference distributions and withhold complete sites before label construction, fitting or model selection. Record the joint predictor support and original observations, including acquisition dates and procedural reconstruction steps. Diagnose saturation using independently measured material; a response mean beyond the historical label maximum alone does not characterize every pixel’s predictor support. Refinement of material labels and their transport can change the estimated stock without changing the general label-inheritance identities.

Sample carbon where deforestation and degradation occur, retaining intermediate stems, large trees, partial crowns, collateral mortality and repeated damage. Apply affected fractions once to the remaining

carbon, then track release, products, dead material, exports and regeneration separately. The relevant association is between carbon-calibration error and the actual source/removal weights. Apply census calibration to both survivor endpoints and the specified mortality stocks, with recruitment and measurement thresholds explicit. The screens in Section 7.5 identify family-specific directions; independent calibration estimates their magnitudes.

### 9.3 Independent reconciliation

Reconstruct physical source/removal histories before assigning them to inventory categories. Aligning managed-land and environmental-change conventions can materially alter an aggregate comparison [23]. On a common carbon-mass boundary,

$$G_{\text{atm}} = E_{\text{fossil}} + F - U - S_{\text{ocean}} + R, \quad (31)$$

fixing all other terms identifies  $F - U$ , while  $(F, U) \mapsto (F + a, U + a)$  preserves that balance. Regional, temporal and tracer observations can constrain the remaining direction [32]. Validation against independently measured components is therefore distinct from forcing an aggregate to close.

## 10 Discussion and conclusions

The framework connects three operations: biological calibration, spatial prediction and annual transfer. A label-trained LiDAR predictor inherits the conditional reference error; changing architectural composition can change a transferred calibration; and release-weighted corrections depend on disturbance and carbon fate. The paired and census operators make source/removal and gain/loss dependence explicit. Their assembly gives a practical validation order: establish material carbon, test spatial transport, and constrain annual exchange with independent observations.

The physical comparison also gives a definite result. Under the illustrative material properties, low-cylinder configurations yield less carbon than the initial class-5 target. Attaining that target requires larger joint basal-area, height and crown-volume products, different material properties, or a revised map. However, a mean crown-mass fraction is neither a whole-tree volume ratio nor a class-wide upper bound. The harvest and TLS literature supports architectural variability and correction of individual-tree estimators; it does not establish the representative distribution needed to validate the initial class mean or the joint bound needed to reject it physically. That magnitude remains unverified, rather than established by the general allometric-bias mechanism.

The empirical claims have distinct scales. Organ measurements and tissue analyses support the original reconstruction chain at their documented domains. The source-table correction changes printed class assignments, leaving carbon stocks unchanged. The support-cap calculation quantifies a numerical operation, not the error of the entire map. All stem sizes enter source populations through abundance, affected fractions and subsequent fates; a sparse large-tree sample cannot establish that distribution.

Figure 2 expresses three conditional consequences of improved forest-carbon measurement: a larger source total when other components are retained, a larger forest contribution when the total and removals are retained, or a joint revision of sources and removals. These alternatives remain analytically valid and empirically testable. A leading forest-loss contribution requires the specified comparison among annual sources; it is neither proved nor universally excluded by the standing-stock estimate. Independent fossil and atmospheric constraints restrict the magnitude and history of any reattribution. The specific two-source net-ranking reversal calculated here lacks the required fossil evidence, while persistent unoffset source additions must fit multiyear balances. Paired correction is one admissible structure, not compulsory cancellation: differential errors act on the source–removal sum. Continued calibration can therefore refine both the estimated carbon and its annual consequences without fixing their outcome in advance.

The ranking implication is affirmative and conditional: when a supported correction makes the defined land-use-and-forestry contribution exceed the largest retained peer sector, it is the largest compartment of that account, even after the increased total is used to renormalize every share. Figure 2 illustrates this property under the explicitly stated 2010 Ecofys grouping; the contemporary net-LULUCF account and the combined energy supersector are different comparisons.

## Data and code availability

The V7.5 companion author archive contains the article, technical supplement, figure plates, transcribed inputs and executable analyses, with an edition-specific SHA-256 manifest. Numerical verification regenerates 76 outputs. Public editions are maintained separately in the [companion research repository](#); the existing V7.0 deposit has its own publication manifest, while this revision is supplied as a companion archive. Editable document sources are excluded from the public repository distribution. Original-record limitations and capture-route fields are documented in `SOURCE_AVAILABILITY.md` and `data/capture_route_schema.json`. No archival DOI has been assigned to this revision.

## Author contribution

Henry Arellano-Peña is the sole author and is responsible for the analysis, interpretation and presentation.

## Competing interests

Henry Arellano-Peña owns the methodologies discussed and is founder and director of NEBIOT S.A.S., the sole company affiliated with this work. NEB001 provides the vegetation-mapping framework and NEB002 the architecture, material-analysis and calibration chain underlying the initial arXiv reassessment [1, 2]. These relationships are disclosed; the author declares no other competing interests.

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## Use of generative artificial intelligence

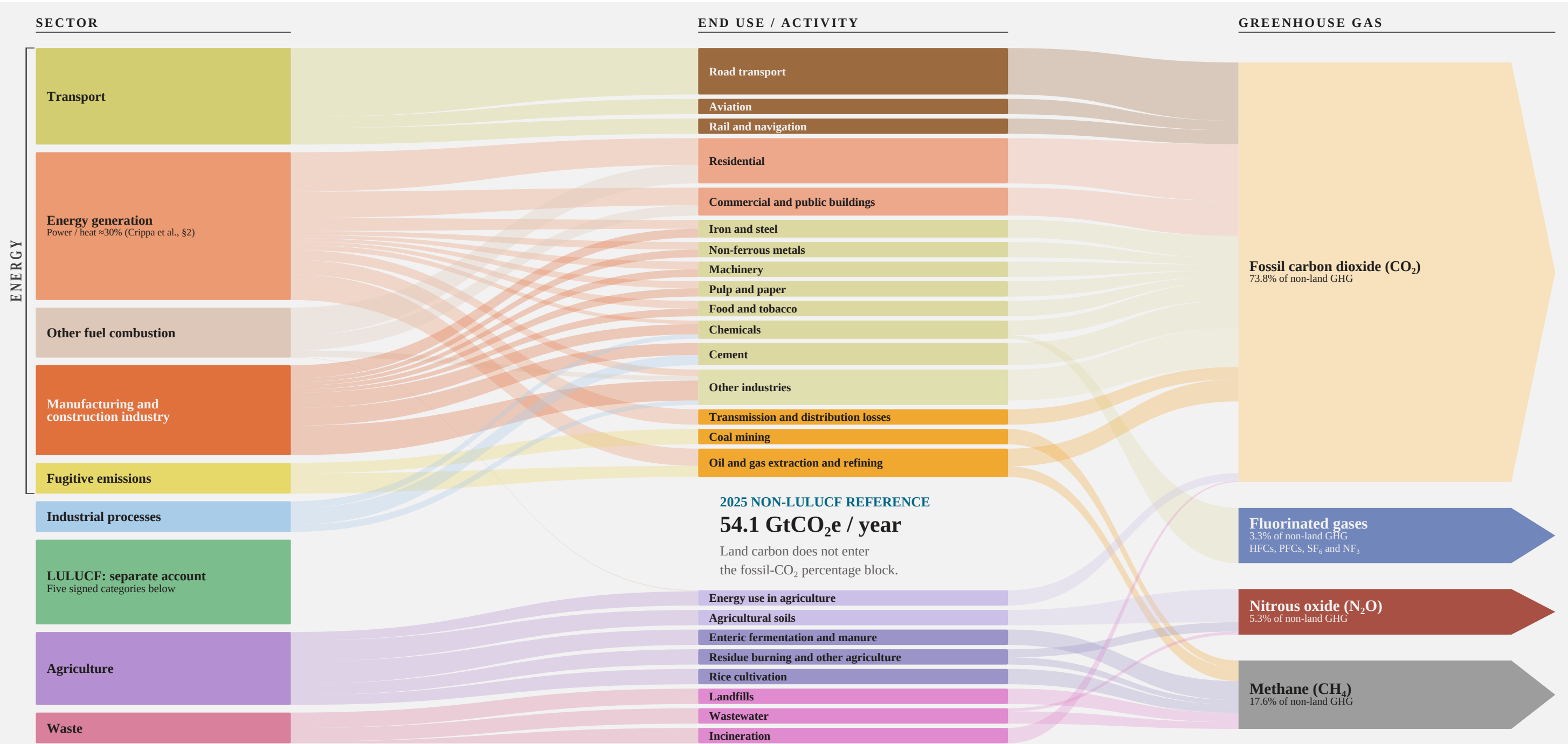
ChatGPT (OpenAI) assisted with drafting, mathematical exploration, source discovery, code and LaTeX preparation. Claude (Anthropic) generated critical-review material used during manuscript development. Model outputs were assessed against the identified sources and analytical derivations rather than treated as empirical evidence. Scientific responsibility remains with the named author.

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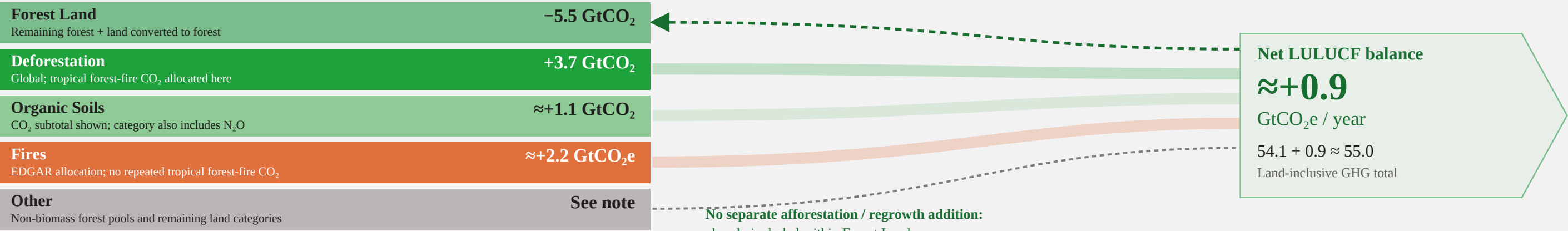
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THE SIGNED LAND ACCOUNT

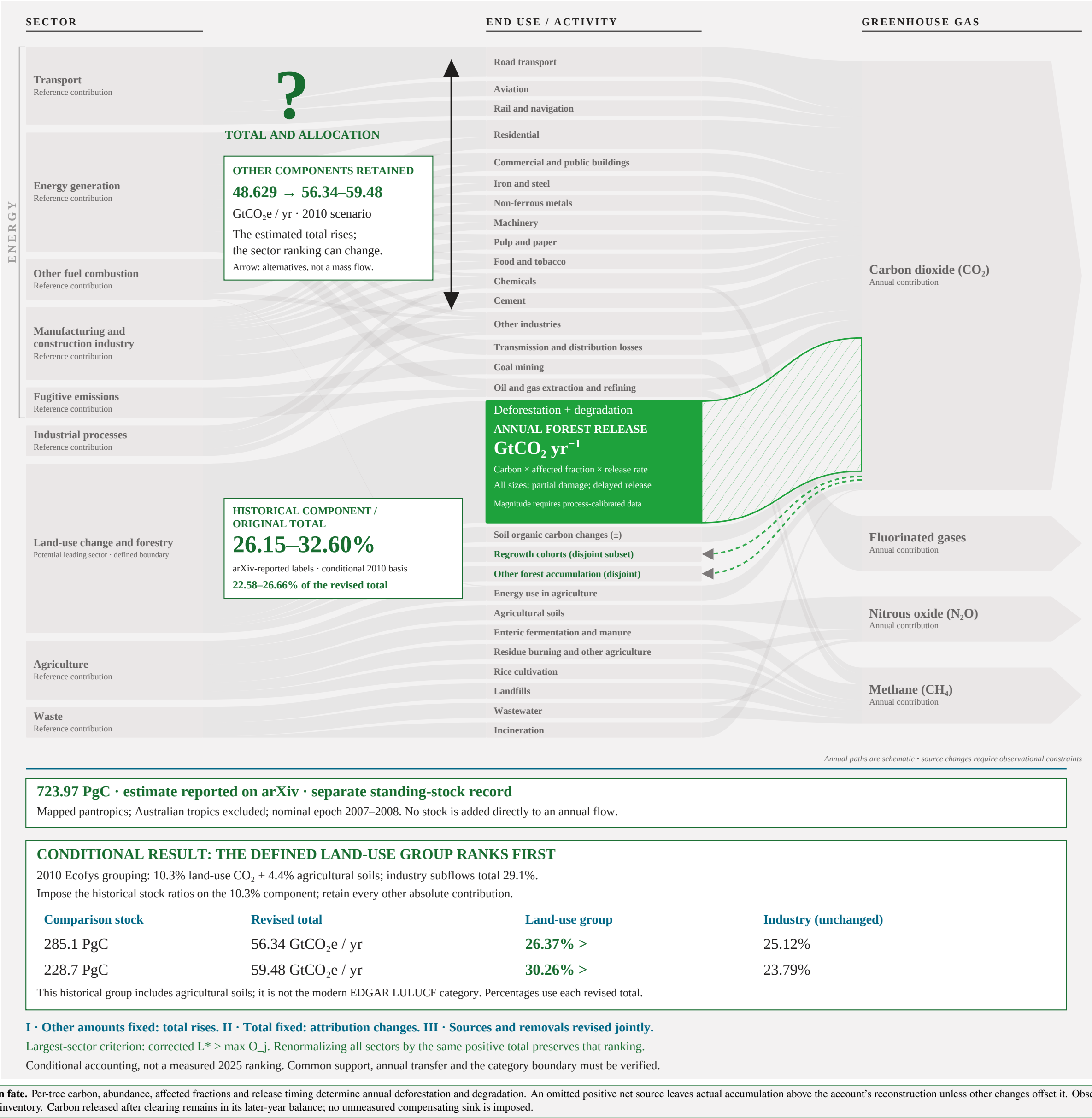
EDGAR 2026 · Section 3, Figure 6 and Annex 2 · 2025 estimates



Other / rounding remainder: ≈-0.6 GtCO<sub>2</sub>e is a closure diagnostic, not an independently extracted category total.

**Disjoint land categories, separate gas denominator.** Forest Land already includes land converted to forest; no extra afforestation or regrowth subtotal is added. Deforestation includes tropical forest-fire CO<sub>2</sub>, which is excluded from Fires; tropical fire non-CO<sub>2</sub> gases remain in Fires. Organic Soils’ quoted 1.1 is CO<sub>2</sub> only. The diagnostic remainder 0.9 − [3.7 − 5.5 + 2.2 + 1.1] ≈ −0.6 also absorbs omitted subcomponents and rounding; it is not a separately extracted Other estimate. Source: Crippa et al. (2026), booklet Sections 2–3 and Annex 2, doi:10.2760/7717504 (CC BY 4.0).

**Figure 1: The reported 2025 account and its boundaries.** The sector–activity–gas organization is adapted from WRI (2005) and Ecofys (2013), with author-created vector artwork and a separate five-category signed land account. Intermediate activity nodes describe pathways, rather than EDGAR’s official sector list. Gas shares refer only to EDGAR’s 54.1 GtCO<sub>2</sub>e non-LULUCF inventory using AR5 GWP<sub>100</sub>; 73.8% is fossil CO<sub>2</sub>. Land emissions therefore do not enter that gas block. The reverse dashed link denotes Forest Land’s net removal. Forest/deforestation estimates use a global Tier 1 approach with default parameters and estimated activity data beyond the latest input maps, not independently remeasured 2025 carbon densities. Paths and all graphical areas are schematic, not a balanced quantitative Sankey. Electricity-attribution paths must not be counted again as direct emissions.



**All stem sizes and explicit carbon fate.** Per-tree carbon, abundance, affected fractions and release timing determine annual deforestation and degradation. An omitted positive net source leaves actual accumulation above the account’s reconstruction unless other changes offset it. Observed air can already contain emissions missed by a bottom-up inventory. Carbon released after clearing remains in its later-year balance; no unmeasured compensating sink is imposed.

**Figure 2: The numerical second interpretation and its conditional largest-sector inference.** The question mark and double-headed arrow concern the estimated total and sector allocation, not an established error in a named energy source. The historical annotation restores the arXiv-reported 26.15–32.60% range; nearest rounding from displayed inputs gives 26.16–32.61%. Under the imposed historical annual multiplier and retained other absolute contributions, totals become 56.34–59.48 GtCO<sub>2</sub>e yr<sup>-1</sup>. In Ecofys’s 2010 grouping, adding the retained 4.4% agricultural-soils component makes its land-use group exceed the 29.1% industry subflow sum at both endpoints, before and after renormalization. That grouping differs from modern EDGAR LULUCF and from the conceptual sector layout above; the numerical card is a separately defined historical example, not a measured 2025 allocation. The component alone ranks above industry only at the upper endpoint. Supplementary Note S3 states the transfer, source-provenance and denominator assumptions. The 723.97 PgC stock remains separate from the annual network. All widths and arrows are schematic; the energy supersector is not a single peer category in the ranking. Layout adapted from WRI (2005) and Ecofys (2013); source interpretation: Arellano-P. and Rangel-Ch. (2015/2016).

CARBON ACCURACY STARTS WITH THE REFERENCE

Spatial coverage measures where; a validated material conversion determines how much.

A • PROXY-LABEL PATHWAY

Reduced predictor measurements

Diameter, canopy height, nominal density and context.  
Organ architecture may remain only implicitly represented.



Allometric biomass / carbon labels

A fitted response summarizes its calibration population.  
Transport can fail when architecture or composition changes.



Remote-sensing calibration

LiDAR or optical metrics are fitted to reference labels.  
Accurate prediction of those labels is not independent mass validation.



Expanded spatial prediction

More coverage can reduce sampling or mapping uncertainty.  
A shared conditional carbon-scale bias can remain.

INHERITANCE IDENTITY • IDEAL CONDITIONAL PREDICTION

$$E[\text{label} \mid \text{LiDAR}] - E[\text{true carbon} \mid \text{LiDAR}] = E[\text{reference-label bias} \mid \text{LiDAR}]$$

Conditional mean-zero residual noise is assumed. This is a mechanism, not an empirical global correction factor.

Initial estimate reported on arXiv

723.97 PgC · architecture-informed calibration  
Nominal map epoch 2007–2008; Australian tropics excluded.  
221 reconstructed trees; 2,232 tissue samples reported.  
Successor calibration in development; no new total reported.

B • ARCHITECTURE-INFORMED PATHWAY

Measured organs and material

Stems, branches, bark and other explicit compartments.  
All stem sizes; biological volume, not saleable timber only.



Carbon from geometry and tissue

Carbon =  $\Sigma$  (organ volume × basic density × carbon fraction).  
Direct laser reconstruction may help measure woody volume.



Independent calibration and holdouts

Validate material geometry, density, chemistry and sampling.  
Keep reference sites separate from model fitting.



Spatial extension with traceable targets

LiDAR can extend a separately validated carbon reference.  
Check transfer, omitted structures and regional extrapolation.

From carbon measurement to annual accounts

All sizes × abundance × affected fractions × release histories.  
Deforestation, degradation, regrowth and other removals.  
Uncounted net release raises reconstructed accumulation.  
A retained total constrains reattribution among sources.

NOT EVERY LiDAR PRODUCT USES THE SAME MEASUREMENT CHAIN

Disney et al. (2018); Calders et al. (2022); Wilkes et al. (2023): independent TLS reconstruction pathways.

Measure architecture and organ allocation to identify the source of allometric calibration error.

Refining calibration can improve the stock estimate without changing the distinction between reference accuracy and coverage.

Sources: Arellano-P. & Rangel-Ch. (2016); Goodman et al. (2014); Calders et al. (2015); GEDI L4A V2 documentation.

Operational diagram: no graphical width represents a measured carbon share. Henry Arellano-Peña.

**The calibration distinction.** A remote-sensing product fitted to allometric reference labels can inherit their conditional bias. Direct reconstruction of biological material with measured tissue properties provides a different reference pathway, which can itself use terrestrial laser measurements. Independent reference tests are required for both approaches; greater coverage alone does not determine absolute carbon accuracy.

**Figure 3: Two measurement-to-map pathways and their annual-accounting consequences.** The left pathway extends a calibrated proxy; the right begins with organ geometry and matched material properties. Architecture and tissue variation identify mechanisms by which an allometric predictor can depart from actual biological carbon. The conditional-expectation identity assumes a source distribution and mean-zero residual noise conditional on the sensor variables; a real learner also has fitting and transfer errors. The initial reconstruction and physical tests are reported in the arXiv study of Arellano-P. and Rangel-Ch. (2015/2016). Crown-related biomass discrepancies are independently illustrated by Goodman et al. (2014). Disney et al. (2018), Calders et al. (2015, 2022) and Wilkes et al. (2023) document terrestrial laser material reconstruction and segmentation, while the GEDI L4A Version 2 guide documents a model linking waveform height metrics and biomass reference data. These examples retain their study domains and establish no universal multiplier. Total material volume includes nonmerchantable woody organs; belowground and dead pools remain separately defined. The 723.97 PgC value is the initial estimate reported on arXiv. Its annual implications require affected populations, release histories and removals. Neither larger spatial coverage nor numerical reproduction of calibration labels is equivalent to independent carbon-mass validation. Original diagram: Henry Arellano-Peña.